

BULGAN, MongoliaWMO: **442390**Lat: **48.8182N**Lon: **103.5190E**Elev: **1208**StdP: **87.63**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-31.3	-36.8	0.1	-33.0	-34.3	0.2	-30.9	8.7	-12.0	7.4	-14.6	1.6	290	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	29.4	17.1	27.2	16.3	25.3	15.7	19.3	25.9	18.1	24.1	17.1	22.8	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	14.2	22.1	16.0	13.2	20.9	15.0	12.3	20.0	60.8	26.1	56.6	24.2	53.1	22.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.3	6.3	-35.7	32.7	3.1	2.5	-37.9	34.4	-39.7	35.9	-41.4	37.3	-43.7	39.1		
(5)			-35.7	21.3	3.0	2.6	-37.9	23.2	-39.6	24.7	-41.3	26.1	-43.5	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-20.4	-16.6	-6.8	3.0	9.2	15.4	17.8	15.5	9.1	0.7	-10.6	-18.0
	DBStd	14.28	5.45	6.02	6.67	4.79	4.51	3.69	2.98	3.11	4.23	4.69	6.76	5.59
	HDD10.0	4336	943	744	521	213	70	4	0	2	64	288	619	868
	HDD18.3	6760	1201	977	779	459	284	101	44	97	276	546	869	1126
	CDD10.0	667	0	0	0	3	45	166	243	173	38	0	0	0
	CDD18.3	50	0	0	0	0	1	12	29	9	0	0	0	0
	CDH23.3	894	0	0	0	1	59	269	384	164	16	0	0	0
	CDH26.7	234	0	0	0	0	10	68	116	39	2	0	0	0
Wind	WSAvg	2.4	1.8	2.0	2.6	3.1	3.2	2.9	2.5	2.4	2.6	2.4	2.0	1.9
Precipitation	PrecAvg	308	1	3	4	12	22	56	100	74	30	9	5	2
	PrecMax	502	9	11	30	68	82	147	217	150	91	40	19	13
	PrecMin	141	0	0	0	0	0	5	14	2	0	0	0	0
	PrecStd	84	2	3	5	11	17	29	51	36	20	8	4	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.0	2.0	14.5	21.8	28.4	31.5	32.9	30.5	25.3	17.8	9.3	-1.8
		MCWB	-5.3	-1.7	5.5	9.8	13.7	16.5	19.6	17.5	14.1	8.8	3.5	-4.3
	2%	DB	-7.0	-1.1	9.9	18.2	25.2	28.6	29.8	27.2	22.6	14.6	5.7	-5.2
		MCWB	-8.8	-4.1	3.1	8.2	12.8	16.2	18.4	16.8	13.3	6.9	1.0	-7.1
	5%	DB	-9.7	-4.1	6.5	15.5	22.6	26.4	27.4	24.9	20.3	11.9	2.3	-7.4
		MCWB	-11.0	-6.2	1.2	6.8	11.6	15.4	17.5	16.1	12.0	5.2	-1.1	-8.9
	10%	DB	-12.0	-7.0	3.6	12.9	20.1	24.3	25.1	22.7	17.7	9.2	-0.8	-9.8
		MCWB	-13.0	-8.4	-0.5	5.5	10.3	14.8	16.8	15.7	10.8	3.9	-3.1	-10.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.2	-1.2	6.1	11.0	16.2	19.1	22.0	20.0	16.3	9.8	4.0	-4.1
		MCDB	-3.2	1.6	13.6	19.2	25.3	27.8	30.2	26.1	21.9	16.5	8.8	-1.9
	2%	WB	-8.6	-3.9	3.5	8.8	14.1	17.7	20.2	18.5	14.4	7.3	1.1	-6.8
		MCDB	-7.1	-1.2	9.2	16.7	21.6	25.3	27.4	24.0	20.6	13.6	5.1	-5.2
	5%	WB	-11.0	-6.1	1.3	7.2	12.4	16.7	18.8	17.3	12.8	5.6	-0.9	-8.8
		MCDB	-9.8	-4.1	6.2	14.6	20.6	23.9	24.8	22.7	19.0	11.2	1.9	-7.3
	10%	WB	-13.0	-8.4	-0.4	5.7	10.9	15.6	17.7	16.2	11.4	4.0	-3.1	-10.9
		MCDB	-11.9	-6.9	3.4	12.3	18.8	22.4	23.1	21.6	16.8	9.0	-0.7	-9.8
Mean Daily Temperature Range	5% DB	MDBR	11.3	14.2	15.0	14.4	15.2	13.8	11.8	12.4	14.5	14.0	12.1	10.6
		MCDBR	13.6	17.0	17.9	19.4	21.0	18.5	16.7	17.0	18.8	18.1	14.9	12.5
		MCWBR	12.1	14.5	11.9	11.1	10.8	8.5	7.4	8.1	10.2	11.1	10.8	11.1
	5% WB	MCDBR	13.5	16.9	17.1	18.1	18.5	15.6	13.8	14.1	16.7	17.5	14.3	12.6
		MCWBR	12.0	14.4	11.7	11.1	10.6	8.2	7.2	7.8	9.7	11.2	10.7	11.1
Clear-Sky Solar Irradiance	taub		0.241	0.264	0.310	0.351	0.381	0.386	0.374	0.349	0.307	0.286	0.262	0.243
	taud		2.337	2.311	2.262	2.260	2.208	2.299	2.383	2.429	2.491	2.475	2.384	2.315
	Ebn at Noon		845	900	902	895	877	874	880	889	902	867	807	785
	Edh at Noon		67	88	111	124	136	126	114	104	88	75	64	60
All-Sky Solar Radiation	RadAvg		1.77	2.90	4.31	5.47	6.41	6.50	6.13	5.37	4.50	3.02	1.84	1.37
	RadStd		0.08	0.15	0.17	0.26	0.33	0.43	0.32	0.29	0.19	0.12	0.09	0.08